

SAGI AMANGELDI

Tokyo, Japan

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Computer Vision Engineer with 3+ years of experience in **3D point cloud processing**, **sensor calibration**, and **algorithm optimization**. Currently working on SLAM and LiDAR-based perception for robotics and industrial scanning.

EXPERIENCE

3D Computer Vision Engineer

July 2025 - now

VITOM Inc., SLAM Team

Tokyo, Japan

- Built the multi-sensor time synchronization layer: PTP across LiDAR, IMU, and camera with hardware/software timestamping, achieving sub-ms camera–LiDAR and ~1 ms IMU–LiDAR precision.
- Replaced the production dynamic-object removal system across all scanner models; optimized memory usage for 64 GB machines, enabling processing of up to 3.5 billion points.
- Developed a colorization pipeline that projects wide-angle RGB imagery onto SLAM-reconstructed LiDAR maps. 1–2 px projection accuracy at 1920×1440 .
- Rebuilt the scanner web app from legacy PHP to Flask with a modular JS frontend.
- Led the port of the SLAM codebase from Python to C++ and the move off Open3D onto in-house geometry.

Computer Vision Engineer

October 2023 - July 2025

Korea Institute of Industrial Technology, 3D Printing Manufacturing Process Center

Ulsan, South Korea

- Built an automated point-cloud inspection pipeline for shipbuilding components covering segmentation, CAD-to-scan registration, and deviation analysis. Delivered the pipeline in joint research with Hyundai Heavy Industries, who built on the algorithms.
- Developed an LLM-assisted search platform for 350+ industrial CAD models using natural-language queries and interactive 3D visualization, running on-premise to keep partner data in their environment.

Software Engineer

December 2022 - October 2023

K-Labs, Software Development Team

Ulsan, South Korea

- Converted a largely manual scan-to-CAD pipeline to partial automation, cutting operator time per conversion.
- Integrated point-cloud post-processing and denoising into internal engineering tools.

SKILLS

Programming: Python, C++, JavaScript

3D Vision & Point Clouds: Open3D, PCL, OpenCV, PyVista, Trimesh, Three.js, CloudCompare, MeshLab

Robotics & SLAM: SLAM, VSLAM, sensor fusion, LiDAR–IMU–Camera calibration, 3D reconstruction, ROS 2

Tools & Platforms: Linux, Docker, PyTorch

Languages: English (advanced), Kazakh (native), Russian (bilingual), Turkish (B2), Korean (TOPIK 2)

EDUCATION

University of Ulsan

Ulsan, South Korea

Master of Science in AI and Computer Engineering, GPA: 4.1/4.5

September 2023 - August 2025

Thesis: Automated Noise Removal and CAD-to-Scan Comparison for Ship Component Inspection Using Terrestrial LiDAR Scanning 📎

Nazarbayev University

Astana, Kazakhstan

Bachelor of Science in Computer Science, GPA: 3.2/4.0

August 2018 - June 2022

POSTERS AND PRESENTATIONS

- **S.Amangeldi**, S.B.Park, J.H.Ha, Y.K.Kwon, D.H.Kim "AI-Assisted Search for Industrial Components Using a Local Database and Large Language Models", International Conference on Precision Engineering and Sustainable Manufacturing 2025 (PRESM2025), Chiang Mai, Thailand, July 2025 (**poster**)
- **S.Amangeldi**, N.T.Phan, A.Tleuliyev, Y.K.Kwon, D.H.Kim "Comparative Analysis of Point Cloud Scans and 3D CAD Data for Dimensional Inspection in Shipbuilding Components", Korean Society for Precision Engineering Conference (KSPE), Jeju, South Korea, May 2024 📎 (**poster**)
- Co-author on three further KSPE 2024 papers, on additive manufacturing for shipbuilding production, supplier selection via genetic algorithms, and audio representations synthesized from 3D models.